

Hyounghoo Kim

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<https://hyounghook.github.io>



INTERESTS

My research designs **Database Systems** for **Hardware Accelerators** (GPU, PIM, ...).

In general, I am interested in using the Accelerators to re-design Data-Intensive Software Systems.

EDUCATION

- **Carnegie Mellon University**, Pittsburgh, Pennsylvania 2023 - Present
Ph.D. Student in Computer Science
Advisor: Phillip B. Gibbons; closely worked with Andrew Pavlo
- **Seoul National University**, Seoul, Korea 2017 - 2023
B.S. in Electrical and Computer Engineering
Advisor: Jangwoo Kim; closely worked with Byung-Gon Chun
GPA: 4.28/4.3 (Summa Cum Laude)
The period includes two years of mandatory military service in South Korea.

PUBLICATIONS

- Hyounghoo Kim, Yiwei Zhao, Andrew Pavlo, Phillip B. Gibbons
No Cap, This Memory Slaps: Breaking Through the Memory Wall of Transactional Database Systems with Processing-in-Memory
Proceedings of the VLDB Endowment, 2025 [Slides] [Poster] [Blog]
- Taebum Kim, Hyounghoo Kim, Gyeong-In Yu, Byung-Gon Chun
BPipe: Memory-Balanced Pipeline Parallelism for Training Large Language Models
International Conference on Machine Learning (ICML), 2023 (Oral Presentation)

EXPERIENCES

- **NVIDIA**, Santa Clara, California Summer 2026
Developer Technology Intern
– GPU Database
- **Microsoft Research (MSR)**, Redmond, Washington Summer 2024, 2025
Research Intern
– GPU Database
– GPU Vector Index
- **Parallel Data Lab & CMU Database Group**, Pittsburgh, Pennsylvania 2023 - Present
Graduate Research Assistant
– (In Progress) Practical GPU OLTP System
– OLTPim: Fast and efficient OLTP on Processing-in-Memory (VLDB 2025)
- **FriendlyAI**, Seoul, Korea 2022 - 2023
Research Intern
– BPipe: Accelerating LLM training by rebalancing memory utilizations (ICML 2023)
- **High Performance Computer System Lab**, Seoul, Korea 2021
Undergraduate Intern
– GPUDiag: Modeling GPU microarchitecture using automated microbenchmarks [Report]
- **Geolux**, Seoul, Korea 2017 - 2018
Software Engineering Intern
– Training AI models to detect potholes from driveway videos

HONORS

- Microsoft PhD Fellowship 2026
- Graduate Fellowship in the School of Computer Science 2025 - 2026
- Northrop Grumman Fellowship - Computer Science 2024 - 2025
- Overseas PhD Scholarship, Korea Foundation for Advanced Studies (KFAS) 2023 - 2028
- Summa Cum Laude, College of Engineering at Seoul National University 2023
- The Presidential Science Scholarship, Korea Student Aid Foundation (KOSAF) 2017 - 2023
- Gold Medal, International Physics Olympiad (IPhO) 2016
- Silver Prize, Samsung Humantech Paper Award 2016

PROJECTS

- Query execution engine for OLAP database systems *Spring 2024*
- Cache simulator for x64 binaries using pintool *Fall 2023*
- Linux kernel hacking to implement custom scheduler, lock, and file system *Spring 2022*
- Compiler frontend for custom grammar rules using lex and yacc *Fall 2021*
- CNN accelerator that can process conv, fc, and maxpool using Verilog and FPGA *Fall 2021*
- IoT system in the billiards ball that evaluates the cueing accuracy 2018
- 3D territory game that adds 3D graphics to the given game logic *Spring 2018*
- Robotic arm that mimics human arm movement 2017
- Robotic arm using thermally-driven super-coiled-nylon artificial muscles 2015 - 2016

TEACHING

- Teaching Assistant, 15-445 "Intro Database Systems", Carnegie Mellon University *Spring 2025*
- Teaching Assistant, "Operating Systems", Seoul National University *Spring 2023*

SKILLS

- C, C++, Python, CUDA, Verilog, Linux Kernel, SQL, PyTorch, ZSim
- Computer Architecture, Database Systems, High-Performance Computing, Machine Learning Systems, Operating Systems